

	STUDENT NAME			COURSE NAME	AUTOMOTIVE DESIGN- CATIA
	REGISTER. NO			TOTAL HOURS	200 HRS
	MOBILE.NO			JOINING DATE	
	CLASS TIMING			STAFF NAME	

SESSION	DATE & HOURS	MODULES	HOURS	STUDENT SIGN	STAFF SIGN
01 SESSION	DATE : _____	Introduction • Understanding CATIA Concepts • Understanding Design Indent • Understanding Workbenches • CATIA application • CAD,CAM,CAE • Understanding Parametric Concepts • Using the CATIA Interface • Understanding the Main Interface • Menu bar,Command bar • Sketcher workbenches • Specifications tree /Model tree • Compass,Planes,Axis • Hide & Show • Setting the Graphics properties • Using the CATIA Interface – Part2 • Arranging the Catia Toolbars • Coordinate system • Geometry • Mouse operations in Catia • Normal View • View tools- ZOOM ,PAN,ROTATE • View orientation-Quick view • Visual styles,View modes • Multiviews ,Fit all in • Opening,Closing and Saving Files • Exit workbench • Managing Files in CATIA windows			
	IN TIME : _____ OUT TIME : _____				
02 SESSION	DATE : _____	Using Sketcher Tools • Geometry operations –Relimitation toolbar Filet,Chamfer,Trim,extend,break,close arc,complement,quick trim • 3D Geometry- Project 3D elements Intersect 3D elements Project 3D silhouette Edges • Transformations Toolbars Mirror,Translate,Rotate, scale,Offset,symmetry			
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03 SESSION	DATE : _____	SKETCH • Creating Sketcher Geometry • Reviewing Sketch Tools- • Snap,construction line • Geometry creation using Profile - • Sketching Lines, Centerlines • Points,Circles, Arcs • Predefined profiles,spline • Using Geometric constraints • Dimensional constraints • Constraints defined in dialog box • Modifying Dimensions within Sketcher • Conflicts-Sketch solving status • Sketch Analysis • Understanding Construction Geometry Theory			
	IN TIME : _____				
	OUT TIME : _____				
04 SESSION	DATE : _____	Part • Introduction to the CATIA Basic Modeling Process • Sketch based feature • Creating Solid Pad • Multipad • Creating Pocket • Multi Pocket • Stiffener • Solid combine			
	IN TIME : _____				
	OUT TIME : _____				
05 SESSION	DATE : _____	Part • Creating Rounds and Chamfers • Creating Rounds by Selecting Edges • face to face fillet • tri tangent Fillet • Chordal Fillet • Variable radius fillet • Analyzing Advanced Chamfer • Dimensioning Schemes • length /length • length /Angle			
	IN TIME : _____				
	OUT TIME : _____				

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06 SESSION	DATE : _____	SKETCH • Creating Holes and Threads • Common Dashboard Options: Hole Depth • Positioning sketch • Extension • Flat bottom ,V bottom • Creating Radial and Diameter Holes • Types of Holes • Simple,Tapered • Counterbored • CountersunkCounterdrilled • Parameters in Holes • Mode,Angle and depth • Anchor point • Thread definition • Thread depth • Pitch • Metric thin pitch • Left and Right threaded			
	IN TIME : _____				
	OUT TIME : _____				
07 SESSION	DATE : _____	Part • Creating round part • Shaft • Revolve Angle • Revolution axis • Groove • Thick profile			
	IN TIME : _____				
	OUT TIME : _____				
08 SESSION	DATE : _____	Part • Shell, Rib, Slot,Thickness • Slot & Rib Features • Creating Shell Features • Creating Thickness • Creating Draft Features • Thread/Tap			
	IN TIME : _____				
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09 SESSION	DATE : _____	Transformation - Transformation features - Translation - Rotation - Symmetry - Axis to Axis - Mirror features - Mirror tool - Mirroring Selected Features - Moving and Rotating Copied Features - Scaling tool - Affinity tool			
	IN TIME : _____				
10 SESSION	OUT TIME : _____	Pattern - Creating Patterns - Rectangular Patterning - Circular Patterning - User pattern - Anchor point - Axial reference - Crown definition - Instance and angular spacing			
	DATE : _____				
	IN TIME : _____				
	OUT TIME : _____				

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11 SESSION	DATE : _____	Boolean • Multibody concept, negative body concept • Insert BODY • Body in a set • Boolean operation • ADD body • REMOVE body • INTERSECT body • ASSEMBLE • UNION TRIM • REMOVE LUMP • GEOMETRICAL SET • Insert geometrical set • Sketch positioning • Projection • Projection point • MINI PROJECT • 3D modeling Using boolean operation • Selecting and Editing • Renaming Objects Regenerating • Activating and Editing Models • define in work object • Cut,Copy • Paste • Parent/children • Local update			
	IN TIME : _____				
12 SESSION	OUT TIME : _____	• SURFACE BASED FEATURES • Split • Thick surface • Close surface • Sew surface • Annotation • Text,Text with leader • Flag note • Flag note with leader • Understanding Remove face • Replace face definition			
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	OUT TIME : _____				

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13 SESSION	DATE : _____	Assembly • Creating New Assembly Models • PRODUCT STRUCTURE TOOLS • Understanding Assembly Theory • Types of Assembly • Terminology • Part,Component,Product • Existing component • Existing component with positioning • Assembly Manipulation • Move • Snap • Smart move • Update • Reframe on			
	IN TIME : _____				
14 SESSION	OUT TIME : _____	constrations • Assembling with Constraints • Understanding Constraint Theory • Fix • Coincident • Surface contact • Offset,Same,Opposite • Angle,Perpendicular • Parallel,Sector • Fix together • Center graph • Change constraints • Quick constraints • Delete constraints • Reorder constraints • Refresh constraints • Activate/Deactivate constraints • Constraints definition			
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	IN TIME : _____				
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15 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	EXPLODE CLASH - Exploding Assemblies - Exploded view - fixed product - Scenes - Enhanced scenes - scenes - exit scenes - scenes browser - Applicative entities reordering - Clash detection - Stop manipulate on clash - Product to product or part to part clash detection			
16 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	SECTION - Space analysis - Sectioning of assembly - Section plane - section slice - section box - view tool			
17 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Measure - Analyzing Model Units - Mass Properties - Measuring Models - Density,Area - Volume - Centre of gravity - Measure Inertia			
18 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Render - Render tool - Photo studio easy tool - Select scenes - Rendering operations - Define Render Area - Save image - Apply Material			

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19 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	- flexible/Rigid sub assembly - how to reorder the assembly - Graph tree reordering - Reuse pattern - Patterns definition - Generated constraints - Reuse the original component			
20 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	DRAFTING Drawing and Detailing - New drawing creation - New drawing, New Sheet - Sheet style - Paper size setting - Standards - Projections setting - Isometric view, front view - Auxiliary view Assignment - Draft the orthographic drawing from the workbook			
21 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Sections - Offset section view - Aligned section view - Offset section cut - Aligned section cut - how to do sections in Isometric view - Assignment - Draft the Part basics drawing from the workbook			
22 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	Details - Detail view - Detail view profile - Quick detail view - Quick detail view profile			

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23 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	• Break view • Broken view • Breakout view • how to apply break views in drawings • Assignment • Do drafting of Radial assembly • Do drafting of Hook assembly			
24 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	• Dimensioning • Length/Distance dimensions • Angle dimension • Radius dimension • Diameter dimensions • Chamfer dimensions • Thread dimensions • Co ordinate dimensions • Give dimensions to all the drawings in the workbook			
25 SESSION	DATE : _____ IN TIME : _____ OUT TIME : _____	• Annotations • Text • Symbol • Roughness value • welding symbol,weld • Add leader • Bill of materials • Title block • Revision table • Table from CSV • Hole Dimension table • Axis system and table parameters • Ballooning • Number Generation • Center mark • Assignment to create BOM and give annotation to all drawings			

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26 SESSION	DATE : _____	• Tolerancing • Datum feature • Apply all type of datum to the drawings • Geometrical tolerance • Adding GD&T to drawings • MINI PROJECT • Reverse engineering • 3D modeling and drafting (from Industry component) • Industrial drawing drafting			
	IN TIME : _____				
	OUT TIME : _____				
27 SESSION	DATE : _____	Reference Elements Reference Elements • Creating Datum Features: Planes ,Points, Lines and Axis • REFERENCE ELEMENT • Planes Definition • Plane type • Offset from plane • Parallel through point • Angle/Normal to plane • Through three points, • Two lines,Point & line • Through planar Curve • Normal to curve • Tangent to surface ,Equation • Mean Through points • POINT DEFINITION • Point type • Co-ordinates ,On curve • On plane, On surface • Circle/Sphere/Ellipse center • Tangent on curve, Between • LINE DEFINITION • Line type • Point -point, Point-Direction • Angle-Normal to curve • Tangent to curve • Normal to surface, Bisecting • AXIS DEFINITION • Axis to Axis • Axis system definition			
	IN TIME : _____				
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28 SESSION	DATE : _____	• Introduction To Sheet Metal Design • The sheet metal design environment, Wall Types • Creating the first wall • Unbend, Bend Back And Cuts			
	IN TIME : _____				
	OUT TIME : _____				
29 SESSION	DATE : _____	PROJECT • Industrial project • Make 3d model and drafting from an Industry component • Customer drawing • Process drawing			
	IN TIME : _____				
	OUT TIME : _____				

COURSE **COMPLETED**

Staff sign

Student sign